EWB CLOUD LABS WORKBOOKS

AWS VPC PEERING REPORT

Project 01 | Oct/Nov 2026

1. COVER PAGE DETAILS

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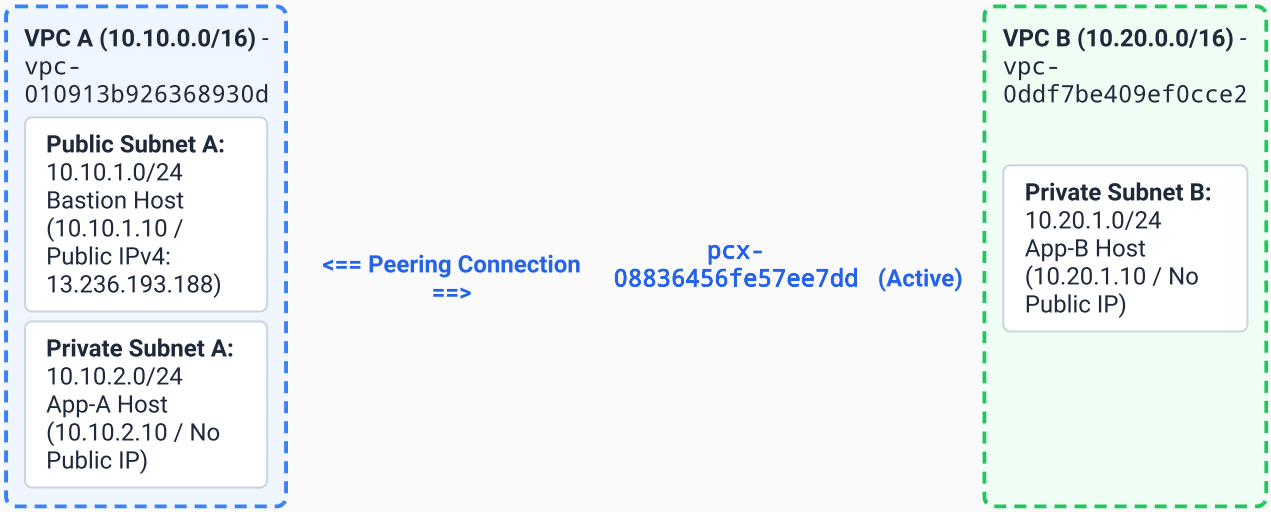
PROJECT TITLE
AWS VPC Peering & Private Cross-VPC Connectivity

AWS REGION
ap-southeast-2 (Sydney)

COMPLETION DATE
November 04, 2026

AWS ACCOUNT ID
362095365428

2. ARCHITECTURE DIAGRAM & RESOURCE INVENTORY



Actual Resource Inventory

Resource Name	Type	Resource ID	IPv4 Address / Subnet
ewb-vpc-a	VPC A	vpc-010913b926368930d	10.10.0.0/16
ewb-vpc-b	VPC B	vpc-0ddf7be409ef0cce2	10.20.0.0/16
ewb-bastion	EC2 (t3.micro)	i-09633ceb529c82864	10.10.1.10 (Pub: 13.236.193.188)
ewb-app-a	EC2 (t3.micro)	i-0b8bcbeaac1597def	10.10.2.10 (Private Only)
ewb-app-b	EC2 (t3.micro)	i-0b572f36249156561	10.20.1.10 (Private Only)

3. SUBNET DETAILS & ROUTE TABLE ASSOCIATIONS

Subnet Name	Subnet ID	CIDR Block	Associated Route Table	Explicit Routes Target
ewb-a-public	`subnet-0e12f74966d7f2a60`	10.10.1.0/24	`rtb-08e7b75d48561d61c` (ewb-rt-a-public)	0.0.0.0/0 → `ewb-igw-a` 10.20.0.0/16 → `pcx-08836456fe57ee7dd`
ewb-a-private	`subnet-072261c3d1d5ea86e`	10.10.2.0/24	`rtb-0868d2c324065e45e` (ewb-rt-a-private)	10.10.0.0/16 → local 10.20.0.0/16 → `pcx-08836456fe57ee7dd`
ewb-b-private	`subnet-0049213dc07da00fa`	10.20.1.0/24	`rtb-049266784cd78f95d` (ewb-rt-b-private)	10.20.0.0/16 → local 10.10.0.0/16 → `pcx-08836456fe57ee7dd`

4. ACTIVE PEERING CONNECTION & SECURITY GROUP RULES

Active Peering Connection: `pcx-08836456fe57ee7dd` (Name: `ewb-a-b-peer`) | Status: ACTIVE
Requester VPC: `vpc-010913b926368930d` (10.10.0.0/16) | Accepter VPC: `vpc-0ddf7be409ef0cce2` (10.20.0.0/16)

Security Group	SG ID	Rule Type	Protocol & Port	Allowed Source
ewb-sg-bastion	`sg-0b3c41142eaed3a91`	SSH	TCP (22)	`223.184.151.102/32` (My Public IP)
ewb-sg-app-a	`sg-0f1661196adf96632`	SSH	TCP (22)	`10.10.1.10/32` (Bastion IP)
		Custom ICMP	Echo Request (Type 8)	`10.20.1.10/32` (App-B IP)
ewb-sg-app-b	`sg-041f8ce2814f09ddf`	SSH	TCP (22)	`10.10.1.10/32` (Bastion IP)
		Custom ICMP	Echo Request (Type 8)	`10.10.2.10/32` (App-A IP)

5. CONNECTIVITY TEST PROOFS

5.1 Baseline Test (Before Peering Route) — Failed Ping

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3119ms
```

5.2 Forward Path (App-A to App-B) — Successful Ping

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.289 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.255 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.299 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.274 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3100ms
```

5.3 Reverse Path (App-B to App-A) & SSH ProxyJump Access

```
PS C:\Users\srishti tiwari> ssh ewb-app-b [ec2-user@ip-10-20-1-10 ~]$ hostname -I 10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10 PING 10.10.2.10 (10.10.2.10) 56(84) bytes of
data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.261 ms 64 bytes from 10.10.2.10:
icmp_seq=2 ttl=127 time=0.233 ms 64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.272 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.245 ms --- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3133ms
```

6. CONTROLLED FAILURE & RECOVERY EXPERIMENT

Test Scenario: Verification of Security Group statefulness and rule isolation by temporarily removing the inbound ICMP rule on App-B.

Step	Action Taken	Observed Output / Status
1. Rule Removal	Removed ICMP Echo Request (Type 8) rule for source `10.10.2.10/32` in `ewb-sg-app-b`.	**100% PACKET LOSS** (Ping times out while SSH remains operational).
2. Rule Restoration	Re-added ICMP Echo Request rule allowing source `10.10.2.10/32` into `ewb-sg-app-b`.	**0% PACKET LOSS** (Ping immediately succeeds with 4/4 replies).

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 # DURING RULE REMOVAL: PING 10.20.1.10
(10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted,
0 received, 100% packet loss, time 3106ms # AFTER RULE RESTORATION: [ec2-user@ip-10-10-2-10
~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from
10.20.1.10: icmp_seq=1 ttl=127 time=0.289 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127
time=0.255 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet
loss, time 3100ms
```

Explanation: AWS VPC Peering creates a network layer path, but security groups act as stateful firewalls at the ENI level. Removing the ICMP rule causes the destination security group (App-B) to drop incoming Echo Requests silently. SSH access via ProxyJump continued working because its separate TCP port 22 rule remained intact.

7. CLEANUP EVIDENCE & KEY LEARNINGS

7.1 Deletion & Teardown Verification

EC2 INSTANCES
Terminated (`i-09633ceb5...`, `i-0b8bcbea...`, `i-0b572f36...`)

VPC PEERING
Deleted (`pcx-08836456fe57ee7dd` + 3/3 routes removed)

ROUTE TABLES
Deleted (`rtb-0868d2c3...`, `rtb-04926678...`, `rtb-08e7b75d...`)

SUBNETS
Deleted (`subnet-0e12f749...`, `subnet-0049213d...`, `subnet-072261c3...`)

SECURITY GROUPS
Deleted (3 SGs removed; only Default SGs remain)

EBS / SNAPSHOTS
0 Active Volumes / 0 Snapshots in Region

Resource Category	Status Banner / Verification Message	State
EC2 Instances	`Successfully initiated termination (deletion) of i-09633ceb529c82864, i-0b8bcbeaac1597def, i-0b572f36249156561`	**SHUTTING-DOWN / TERMINATED**

Resource Category	Status Banner / Verification Message	State
Peering Connection	`You have successfully deleted peering connection pcx-08836456fe57ee7dd (3 of 3 related routes were deleted)`	DELETED
Route Tables	`You have successfully deleted rtb-0868d2c324065e45e/ewb-rt-a-private, rtb-049266784cd78f95d/ewb-rt-b-private, rtb-08e7b75d48561d61c/ewb-rt-a-public`	DELETED
Subnets	`You have successfully deleted subnet-0e12f74966d7f2a60, subnet-0049213dc07da00fa, subnet-072261c3d1d5ea86e`	DELETED
Security Groups	`Successfully deleted 3 security groups`	DELETED
Storage Audit	`You currently have no volumes / snapshots in this region (0/0)`	CLEAN

Console Verification: Account returned to default VPC state (vpc-08c623935b5a637c5). All custom subnets, security groups, route tables, internet gateways, and peering connections have been completely cleaned up.

7.2 What I Learned (Key Takeaways)

Non-Overlapping CIDRs: VPC Peering strictly requires non-overlapping IPv4 CIDRs (10.10.0.0/16 and 10.20.0.0/16). **Bidirectional Routing:** Adding a route in VPC A without a return route in VPC B results in asymmetric traffic loss; both route tables must point to the `pcx` ID. **ProxyJump Administration:** Secure administrative access to private instances without public IPs is best achieved using standard SSH ProxyJump via a hardened Bastion. **Peering Non-Transitivity:** VPC Peering does not support edge routing or transit routing (e.g., VPC B cannot access the Internet through VPC A's IGW).